Davide-Andrei Mureșan

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About Me

Curiosity is what moves me: I want to know how things actually work, to open them up, follow what happens inside, and understand why each piece is there. That is what drew me to engineering.

I stay calm when things go wrong. Instead of panicking, I immediately focus on finding a rational, logical solution. I know how much I still have to learn, and that is exactly why I want to work alongside people with more experience than mine: I learn quickly, I ask when I do not understand something, and I will give my best to help the team bring every task to completion.

Education

MSc Communications Networks & Distributed Systems (RCSD) 2026 – Present
Technical University of Cluj-Napoca, Faculty of Automation and Computer Science *Cluj-Napoca*

Research proposal: **Distributed Pedestrian Crossing-Intent Estimation over V2X Vehicular Networks.** Extends my thesis beyond a single vehicle's sensors: several vehicles and roadside sensors pool what they observe, and the research asks what is worth transmitting and when, given limited bandwidth.

BSc Computer Science and Information Technology 2022 – 2026
Technical University of Cluj-Napoca, Faculty of Automation and Computer Science *Cluj-Napoca*

Relevant coursework: Control Systems (transfer functions, PID control), Computer Networks, Communication Protocols, Data Structures & Algorithms, Operating Systems, Databases, Software Design, Artificial Intelligence.

Experience

IT – Web, Mobile & Industrial Development Intern Jul 2025 – Sept 2025
Tenaris *Cluj-Napoca, Romania*

- Developed features in C# for iControl, the internal platform of the plant used to plan and track production scheduling.
- Designed and implemented an AI assistant guiding users through the platform's functionality, much of which was previously reachable only by asking a colleague who knew where to look, integrated across web, mobile and industrial software.
- Collaborated in an Agile team through the full cycle: refinement, implementation, testing and review.

Projects

Three-Building Campus Network: Design, Configuration and Automation Cisco Packet Tracer, Python, Jinja2, YAML, pytest, CI

A campus network built across three sites: **VLANs** and **Spanning Tree** at Layer 2, **OSPF** at Layer 3, **DHCP**, **WPA2** wireless and a **DMZ** for public services, with every device hardened through separate privilege levels and SSH-only management. Two thirds of the device configurations were **generated from a YAML inventory** through Jinja2 templates and verified by an automated test suite in CI. [GitHub]

SafeCross: Pedestrian Crossing-Intent Estimation Bachelor's Thesis | Python, PyTorch, C++, TensorRT

Driver-assistance system that predicts whether a pedestrian intends to cross. A pipeline of AI models (detection and segmentation, a custom network for body orientation, and a classifier over tracked geometric features) running on both **RGB and thermal** imagery, so it holds up at night as well as in daylight. Body orientation is estimated to **93% accuracy**, and the whole system runs in real time through a C++ and TensorRT deployment that cut detection from 74 ms to **20 ms**.

TalentMatch: AI-Powered CV-to-Job Matching 2nd Place, AI Hackathon by Accesa | Python, Java, Angular, Azure

Web application helping recruiters match CVs against job descriptions, combining semantic similarity with explicit rules. I coordinated the four-person team, built the CV parsing pipeline on the Gemini API, implemented two of the three scoring components, and set up the connection to Azure services. [GitHub]

Communication Protocols on FPGA VHDL, Vivado

Implemented I^2C, SPI and Ethernet from the specification in VHDL: transmitter and receiver for each, driven by finite state machines and verified in simulation. [GitHub]

Certifications

- **Cisco CCNA 1: Introduction to Networks** · Cisco Networking Academy (Ethernet, IP subnetting)
- **Cisco CCNA 2: Switching, Routing & Wireless** · Cisco Networking Academy (STP, EtherChannel)
- **Cambridge Advanced Certificate (CAE)** · English proficiency

Skills

Systems	Linux (Ubuntu, command line), Windows, Git, GitHub Actions, CI pipelines, network interface configuration
Networking	TCP/IP, IPv4/IPv6 subnetting, VLANs, inter-VLAN routing, STP, EtherChannel, HSRP, OSPF, ACLs, NAT/PAT, DHCP, DNS, Cisco IOS CLI
Programming	Python, Java, C/C++, C#, SQL, Bash, VHDL, Assembly x86; Jinja2, YAML, pytest
AI / ML	PyTorch, OpenCV, ONNX, TensorRT; FastAPI, Spring Boot, Angular, .NET
Soft skills	Teamwork and collaboration, public speaking and presentation, explaining technical work to people outside the field
Languages	Romanian, English, Italian